Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building backend systems, ETL workflows, and AI-powered applications, with hands-on experience across LLM APIs, embeddings, evaluation pipelines, and database-backed engineering data workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and ETL-style workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, schema updates, and maintainable release pipelines across cloud-hosted systems and structured data operations.
- Improved **system performance and reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening traceability, debugging, and production support for shared data platforms.
- Supported **multi-tenant data operations** across relational workflows and cloud storage, helping maintain consistent validation, access patterns, and production handoffs for engineering-facing platform features and recurring integration jobs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, ETL workflows, and reliable release coordination across operating environments.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions while strengthening defect isolation, reproducibility, and documentation across recurring build and integration cycles.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and multimodal LLM workflows** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence logic, and model-backed evaluation pipelines to improve reliability and tooling for production users.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading features, documenting fixes and reducing feedback turnaround by **60%** while improving observability, issue isolation, and maintainability across AI-assisted product workflows.
- Tracked **model behavior and workflow outputs** across grading, OCR, and persistence paths, helping turn ad hoc debugging into repeatable evaluation, validation, and support practices for AI-assisted product features.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a **GPT-based Decision Transformer** for offline RL in autonomous driving, improving long-horizon trajectory planning by **18%** in simulation while developing end-to-end Python pipelines across 100K+ trajectory logs.
- Architected a **spatial-temporal encoder** combining HD-map geometry, lane direction, and ego-agent history through multi-head attention, improving closed-loop planning stability by **15%** for autonomous-driving evaluation and validation workflows.

Skills

Languages: TypeScript, Node.js, Python, JavaScript, SQL, Java, C++
AI/LLM: OpenAI-style APIs, Anthropic-style workflows, embeddings, vector search, prompt design, evaluation pipelines, multimodal tooling
Data & APIs: REST APIs, ETL/ELT workflows, orchestration, SQL/NoSQL, PostgreSQL, MongoDB, AWS Glue/S3, Azure SQL
Delivery: Docker, Jenkins, Git/GitHub, CI/CD, debugging, documentation, production support

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **AI-powered document platform** using embeddings, FAISS, Pinecone, and PostgreSQL to support semantic search, summarization, and collaborative workflows across 50K+ records with low-latency retrieval and structured data handling.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data management from a unified interface while working across Java backend and TypeScript frontend modules.

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81%** to **96%** through repeatable evaluation pipelines and structured error analysis.